Article

Periodic Boundary Conditions in Generalized Beam Theory: A Shear-Deformable GBT–Ritz Formulation for the Buckling of Omega-Stringer-Stiffened Composite Panels

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Abstract

Stringer-stiffened composite panels are commonly analysed on one representative cell with periodic boundary conditions. Generalized Beam Theory (GBT) describes the buckling of thin-walled members by cross-section modes, but periodic side edges have not been treated in its cross-section analysis. This paper builds the periodic condition into every family of GBT modes, so that every mode and every truncated basis satisfies it exactly. A shear-deformable GBT–Ritz member analysis of omega-stringer-stiffened panels follows, with laminated Reissner–Mindlin walls, the true stringer offset, a computed pre-buckling state and B-spline amplitude functions; it reduces to classical GBT for uncoupled walls. For 30 cells with simply supported loaded edges, the critical loads agree with periodic Abaqus shell models within 0.57 % on average and 1.25 % at most; with clamped loaded edges within 0.64 % and 1.52 %. Short cells buckle locally. From about six stringer pitches, isotropic and angle-ply cells buckle in a long-wave mode driven by the transverse compression that the constant cell width induces, whereas cross-ply and unsymmetric cells buckle globally from twelve pitches. Abaqus confirms this mode and, for two cells allowed to expand laterally, gives 2.0–2.9 times higher critical loads; the lateral restraint assumed in periodic analyses thus matters for long panels.

Keywords: generalized beam theory; Ritz method; periodic boundary conditions; stiffened panels; omega stringer; laminated composites; buckling

1. Introduction

Stringer-stiffened panels are the standard construction of aircraft fuselage and wing covers, and buckling is often their governing design criterion [1,2]. The omega (hat) stringer offers high bending and torsional stiffness and a wide bonding area [3,4]. Away from its longitudinal edges, a cover with many identical stringers is represented by one repeating cell with periodic boundary conditions [5–7]. The principle goes back to waves in periodic structures [8] and is used in exact strip programs (VIPASA, VICONOPT) [9,10] and finite strip methods [11,12].

At the Institute for Lightweight Engineering and Structural Mechanics (LSM), closed-form models of stiffened composite panels have been developed for blade stringers [5,6] and omega stringers [3,13], including local and column buckling [4,7], shear and combined loading [14,15] and postbuckling [16], and most recently local buckling and postbuckling of omega-stringer panels including the stringer feet [17–19]; see also [20]. These models reduce the panel to a repeating unit with periodic [5–7] or symmetry conditions [18], the latter admitting only modes symmetric about the symmetry lines. They are efficient but idealise the skin–stringer interaction by elastic restraints, separate local and global buckling, assume a uniform pre-buckling stress and include membrane–bending coupling through a reduced bending stiffness [21], as does the physics-informed neural network of the authors [22]. Shell finite element models avoid these idealisations but are expensive and give the mode only as a nodal field.

Generalized Beam Theory (GBT), originated by Schardt [23,24] and developed further by Davies and Leach [25], describes a thin-walled member by cross-section modes with amplitude functions along the axis, which makes it transparent for coupled instabilities [26–29]. Silvestre and Camotim formulated GBT for orthotropic and laminated members [30,31] including shear deformation [32,33]. The cross-section analysis was extended to branched sections [34], reformulated [35,36] with shear modes [37] and implemented in GBTUL 2.0 [38]; closed sections were treated for tubes [39,40], and GBT was applied to fibre-reinforced polymer (FRP) columns [41], stiffened composite panels [42] and stiffened sections [43]. The Ritz method complements GBT in the member analysis [44–46]: Kharghani and Mittelstedt combined both into a GBT–Ritz approach for global, local and distortional buckling of shear-deformable composite beams [47], composite box girders [48] and beams with unsymmetric lay-ups [49,50].

GBT has been developed for members with free longitudinal edges or closed contours, whereas the skin of a panel cell ends at two side edges whose displacements must be equal. This condition has not been formulated in GBT, and it cannot simply be imposed on a truncated set of classical modes: these are computed for an open contour with independent side edges, so that few of their combinations are periodic. Moreover, the omega stringer forms closed cells with the skin, which excludes Vlasov's warping, it is offset from the skin, and the pre-buckling state is not uniform.

The present study closes this gap. Its novelty is the formulation of the periodic condition within the cross-section analysis of GBT, such that every mode, and therefore every truncated basis, satisfies it exactly, together with a shear-deformable GBT–Ritz analysis [50] of omega-stringer-stiffened panels with symmetric and unsymmetric laminates, verified against Abaqus shell models. The mode decomposition reveals, besides the transition from local to global buckling of long cells, a long-wave mode caused by the constant width that the periodic condition imposes. Section 2 derives the formulation, Section 3 verifies it against Abaqus and discusses the results, and Section 4 concludes.

2. Theory and formulations

GBT analyses a thin-walled member in two stages [24,30,36]: the cross-section analysis (Sections 2.1 and 2.2) yields the deformation modes and their stiffness constants (modal tensors); the member analysis (Section 2.3) yields the amplitude functions of the modes along the length and the critical load. The notation is that of Silvestre and Camotim [32], and the member analysis follows the GBT–Ritz approach [47,50]. Classical GBT is extended, and recovered as a special case (Section 2.3.4), by periodic modes, the main contribution (Section 2.2), closed contours without Vlasov's assumptions, Reissner–Mindlin laminated walls and a computed pre-buckling state.

2.1. Cross-section analysis

2.1.1. Periodic cell and displacement field

The cell of the infinitely wide panel (Figure 1b) contains two stringers at the pitch b_s; it has the width $W = 2b_s$ and the length $L = \beta b_s$, with the aspect ratio β (Figure 1a). Two stringers are needed because a cell of one pitch contains only modes that repeat in every bay, not those that alternate from bay to bay (Section 2.2.5). X runs across the panel, Y normal to the skin and x along the stringers; the skin occupies $-W \le X \le 0$, with the side edges s_L ($X = -W$) and s_R ($X = 0$) in the flat middle of a skin bay. Every wall has local axes (x, s, z) (s along its mid-line, z normal) with mid-surface displacements u, v, w [32]. The walls are divided into segments of at most $b_s/25$ [24,36]; for $b_s = 150$ mm the cell has $N = 92$ nodes, 19 walls and 93 segments (Figure 1c).

In GBT, every displacement is a sum of mode profiles (functions of s) times amplitude functions (functions of x) [32],

$$u(x,s) = u_k(s)\,\phi_{k,x}(x) + u_j(s)\,\varphi_j(x), \qquad v(x,s) = v_k(s)\,\phi_k(x), \qquad w(x,s) = w_k(s)\,\phi_k(x), \quad (1)$$

with summation over repeated indices. The conventional modes k deform the cross-section in its plane (v_k, w_k), with a warping u_k tied to $\phi_{k,x}$ as in beam theory; the shear modes j add an independent warping u_j, i.e. mid-surface shear, and include the axial extension $u_j = 1$. The rotations of the Reissner–Mindlin wall normals are described by transverse shear (Mindlin) modes r,

$$\psi_x(x,s) = \xi_r(s)\,\chi_r(x), \qquad \psi_s(x,s) = \zeta_r(s)\,\chi_r(x), \quad (2)$$

with the profiles ξ_r, ζ_r and amplitudes χ_r; in variations, i and h replace k and j.

2.1.2. Strains and constitutive relations

Inserting Eqs. (1) and (2) into the Reissner–Mindlin strains, i.e. membrane strains plus z times the curvatures $\kappa_{xx} = -w_{,xx} + \psi_{x,x}$, $\kappa_{ss} = -w_{,ss} + \psi_{s,s}$, $\kappa_{xs} = -2w_{,xs} + \psi_{x,s} + \psi_{s,x}$, and the transverse shear strains $\gamma_{xz} = \psi_x$, $\gamma_{sz} = \psi_s$, gives

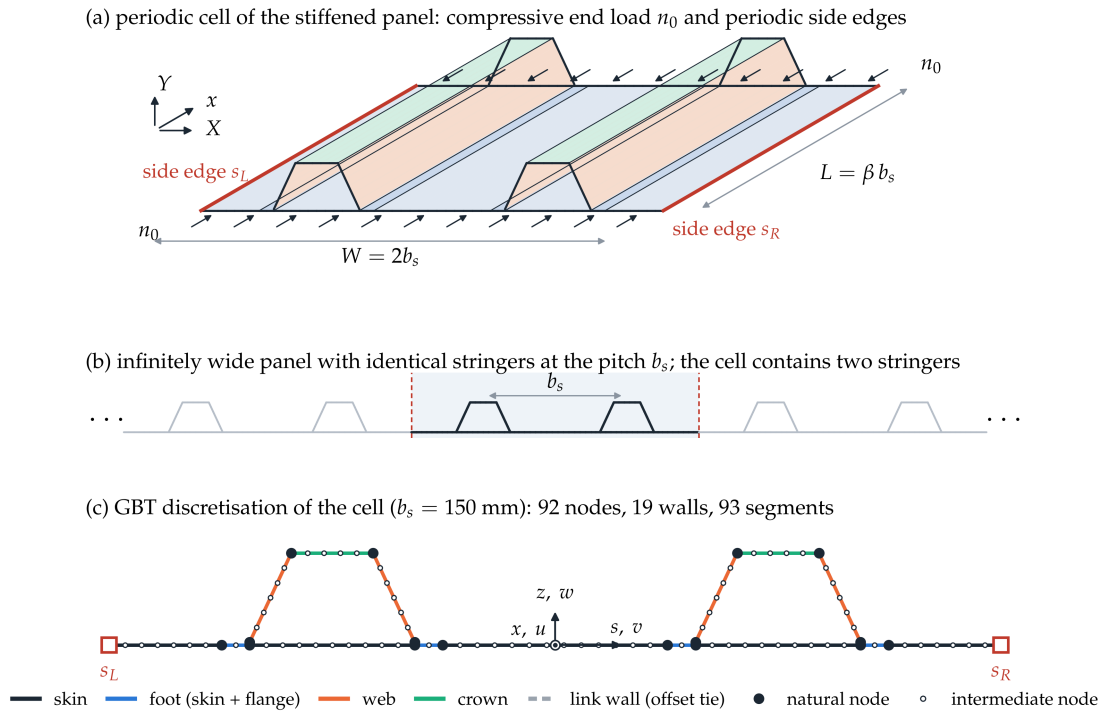


Figure 1. Periodic cell of the omega-stringer-stiffened panel. (a) Cell of width $W = 2b_s$ and length $L = \beta b_s$ with the end load n_0 and the side edges s_L, s_R linked by the periodic condition. (b) Infinitely wide panel and the cell (shaded). (c) GBT discretisation for $b_s = 150$ mm with nodes, wall types and local axes (x, s, z).

$$\begin{aligned} \varepsilon_{xx} &= (u_k - z\,w_k)\,\phi_{k,xx} + u_j\,\varphi_{j,x} + z\,\xi_r\,\chi_{r,x}, \qquad \varepsilon_{ss} = \left(v_{k,s} - z\,w_{k,ss}\right)\phi_k + z\,\zeta_{r,s}\,\chi_r, \\ \gamma_{xs} &= \left(\gamma_k - 2z\,w_{k,s}\right)\phi_{k,x} + u_{j,s}\,\varphi_j + z\left(\xi_{r,s}\,\chi_r + \zeta_r\,\chi_{r,x}\right), \qquad \gamma_{xz} = \xi_r\,\chi_r, \qquad \gamma_{sz} = \zeta_r\,\chi_r, \end{aligned} \tag{3}$$

with the membrane shear strain profile $\gamma_k = u_{k,s} + v_k$. In [32] the conventional modes satisfy Vlasov's assumptions $\gamma_k = 0$ (no membrane shear) and $v_{k,s} = 0$ (no transverse extension) and the walls are Kirchhoff plates ($\xi_r = \zeta_r = 0$); here, transverse extension modes are included and $\gamma_k = 0$ cannot hold in closed contours (Section 2.2.3). The geometric stiffness requires the quadratic membrane strains

$$\varepsilon_{xx}^{NL} = \tfrac{1}{2}\left(u_{,x}^2 + v_{,x}^2 + w_{,x}^2\right), \qquad \varepsilon_{ss}^{NL} = \tfrac{1}{2}\left(u_{,s}^2 + v_{,s}^2 + w_{,s}^2\right), \qquad \gamma_{xs}^{NL} = u_{,x}u_{,s} + v_{,x}v_{,s} + w_{,x}w_{,s}, \tag{4}$$

of which classical GBT retains $\frac{1}{2}(v_{,x}^2 + w_{,x}^2)$ [31,32]. The membrane forces **N**, moments **M** and transverse shear forces **Q** follow from first-order shear deformation laminate theory with a shear correction factor of 5/6 [51–53],

$$\begin{bmatrix} \mathbf{N} \\ \mathbf{M} \\ \mathbf{Q} \end{bmatrix} = \begin{bmatrix} \mathbf{A} & \mathbf{B} & \mathbf{0} \\ \mathbf{B} & \mathbf{D} & \mathbf{0} \\ \mathbf{0} & \mathbf{0} & \mathbf{A}_z \end{bmatrix} \begin{bmatrix} \varepsilon^M \\ \kappa \\ \gamma_z \end{bmatrix}, \tag{5}$$

with the membrane, coupling, bending and transverse shear stiffnesses **A**, **B**, **D**, $\mathbf{A}_z$. All couplings are retained [32,50], and N_{ss} is not set to zero as in beam theories, since the cell restrains the transverse expansion of the skin (Section 3.2).

Two walls need special treatment. A stringer foot (skin plus bonded flange) is one wall whose stiffness is referred to the skin mid-plane, so that it joins the neighbouring skin at the same height. With the stiffnesses $\tilde{\mathbf{A}}$, $\tilde{\mathbf{B}}$, $\tilde{\mathbf{D}}$ of the combined stack about its own mid-plane, at $z_f = t_f/2$ from the skin mid-plane towards the flange, the standard shift of the reference plane ($z = z' + z_f$ in the stiffness integrals) gives

$$\mathbf{A}_f = \tilde{\mathbf{A}}, \qquad \mathbf{B}_f = \tilde{\mathbf{B}} + z_f\,\tilde{\mathbf{A}}, \qquad \mathbf{D}_f = \tilde{\mathbf{D}} + 2z_f\,\tilde{\mathbf{B}} + z_f^2\,\tilde{\mathbf{A}}. \tag{6}$$

The second wall reproduces the offset $c = (t_s + t_f)/2$ between the stringer and skin mid-surfaces (thicknesses t_s, t_f), bridged by a tie in the shell model. Every web root is a node at $Y = c$, joined to the skin node below by a link wall that must transfer displacements and rotation rigidly without adding material; its stiffnesses are multiples of the skin stiffnesses $\mathbf{A}_0$, $\mathbf{D}_0$, $\mathbf{A}_{z0}$,

$$\begin{gathered} A^L_{11} = 10^{-3} A_{0,11}, \quad A^L_{22} = k_L A_{0,22}, \quad A^L_{66} = k_L A_{0,66}, \\ D^L_{11} = 10^{-3} D_{0,11}, \quad D^L_{22} = k_L D_{0,22}, \quad D^L_{66} = 10^{-3} D_{0,66}, \quad \mathbf{A}^L_z = k_L \mathbf{A}_{z0}, \end{gathered} \tag{7}$$

with all other entries zero. The factor $k_L = 10^3$ makes the link rigid in the cross-section plane; the factor 10^{-3} removes its axial, longitudinal bending and twisting stiffness, which a row of rigid links does not have. Both are numerical choices: changing k_L from 10^2 to 10^4 or the small factor from 10^{-2} to 0 changes the critical load by less than 0.01 % (Section 3.7).

2.1.3. Nodal description and mode families

As in modern GBT [36,37], the modes are built from nodal quantities: on every segment p (width b_p, unit tangent $\mathbf{t}_p$, end nodes q, q', local coordinate $\eta \in [0,1]$), u and v vary linearly, and w is the cubic Hermite interpolation of the nodal normal displacements w_q and rotations $\theta_q = w_{,s}$,

$$\begin{gathered} w(\eta) = \mathcal{H}_1(\eta)\, w_q + b_p \mathcal{H}_2(\eta)\, \theta_q + \mathcal{H}_3(\eta)\, w_{q'} + b_p \mathcal{H}_4(\eta)\, \theta_{q'}, \\ \mathcal{H}_1 = 1 - 3\eta^2 + 2\eta^3, \quad \mathcal{H}_2 = \eta - 2\eta^2 + \eta^3, \quad \mathcal{H}_3 = 3\eta^2 - 2\eta^3, \quad \mathcal{H}_4 = -\eta^2 + \eta^3. \end{gathered} \tag{8}$$

This is equivalent to Schardt's nodal moments [24], but makes the periodic conditions linear in the nodal quantities. The in-plane deformation is described by the nodal displacements $\mathbf{U}$ (a vector of the $2N$ components U_X, U_Y of all nodes) and the nodal rotations Θ (N components), and the segment elongations by the matrix $\mathbf{E}_\ell$, $(\mathbf{E}_\ell \mathbf{U})_p = \mathbf{t}_p \cdot (\mathbf{U}_{q'} - \mathbf{U}_q)$. The modes are generated in the usual GBT families:

- **G** (global): rigid-body translations and rotation;
- **D** (distortional): nodal displacements without segment elongation ($\mathbf{E}_\ell \mathbf{U} = \mathbf{0}$), orthogonal to G and L, which move the wall junctions;
- **L** (local): a unit normal displacement of a non-junction node or a unit nodal rotation, i.e. wall bending between junctions;
- **T** (transverse extension): the columns of $\mathbf{E}_\ell^{\mathsf{T}}$, which stretch the segments;
- **S** (shear): the nodal hat functions of the warping, one replaced by the axial mode $u_j = 1$;
- **M** (Mindlin): nodal vectors Ω_q with $\xi_r = \mathbf{t} \cdot \Omega_{q'}$ and nodal values of ζ_r, interpolated linearly.

G, D and L span the null space (kernel) $\ker \mathbf{E}_\ell = \{\mathbf{U} : \mathbf{E}_\ell \mathbf{U} = \mathbf{0}\}$ of fields that elongate no segment, and T spans the range of $\mathbf{E}_\ell^{\mathsf{T}}$ (the space spanned by its columns); by the standard decomposition $\mathbb{R}^{2N} = \ker \mathbf{E}_\ell \oplus \operatorname{range} \mathbf{E}_\ell^{\mathsf{T}}$ of linear algebra, the families thus cover every in-plane deformation, with the rotations in family L. The warping u_k of the G and D modes is not free: classical GBT chooses it such that the membrane shear strain γ_k vanishes [24,54]. In a closed contour this is impossible, so u_k minimises the membrane shear strain energy,

$$\min_{\mathbf{u}} \ \Pi_w(\mathbf{u}) = \sum_p \hat{A}_{66,p}\, b_p \left(\frac{u_{q'} - u_q}{b_p} + v_p\big(\eta = \tfrac{1}{2}\big) \right)^2 \quad \text{subject to} \quad \mathbf{c}_{ax}^{\mathsf{T}} \mathbf{u} = 0, \tag{9}$$

where $\mathbf{u}$ collects the nodal warping values, the bracket is γ_k at the segment mid-point, $\hat{A}_{66} = A_{66} - A_{26}^2/A_{22}$ is the shear stiffness for zero transverse force, and the constraint, with $c_{ax,q} = \sum_{p \ni q} \frac{1}{2} \hat{A}_{11,p} b_p$ and $\hat{A}_{11} = A_{11} - A_{12}^2/A_{22}$, removes the axial resultant (axial extension is the separate mode $u_j = 1$). For an open contour the minimum is zero and Vlasov's warping is recovered; in a closed contour the remaining γ_k is kept in Eq. (3).

2.2. Periodic boundary conditions in the cross-section analysis

2.2.1. Continuous condition and its modal form

As one period of the infinite panel, the cell has side edges that are the same line of the panel, so all displacements and rotations must be equal on them,

$$\mathcal{U}_\alpha(X=-W,\,x)-\mathcal{U}_\alpha(X=0,\,x)=0,\qquad \alpha=1,\dots,6,\qquad 0\le x\le L, \tag{10}$$

where $\mathcal{U}_1$–$\mathcal{U}_3$ are the translations in X, Y, x and $\mathcal{U}_4$–$\mathcal{U}_6$ the rotations about X, Y, x of a shell node; shell models impose Eq. (10) with one equation per pair of side nodes and degree of freedom [5–7]. Table 1 gives the corresponding conditions on the GBT modes.

Table 1. Degrees of freedom of a skin side node and the corresponding modal conditions.

Shell DOF	Meaning at a skin side node	Modal condition
$\mathcal{U}_1$	in-plane displacement U_X	$U_{X,s_L}=U_{X,s_R}$ for every conventional mode, Eq. (12)
$\mathcal{U}_2$	deflection U_Y	$U_{Y,s_L}=U_{Y,s_R}$ for every conventional mode, Eq. (12)
$\mathcal{U}_3$	axial displacement u	$u_k(s_L)=u_k(s_R)$, Eq. (13); $u_j(s_L)=u_j(s_R)$, Eq. (14)
$\mathcal{U}_4$	rotation $-w_{,x}+\psi_x$	follows from $\mathcal{U}_2$ for $w_{,x}$; $\xi_r(s_L)=\xi_r(s_R)$, Eq. (14)
$\mathcal{U}_6$	rotation $\theta+\psi_s$	$\theta_{s_L}=\theta_{s_R}$, Eq. (12); $\zeta_r(s_L)=\zeta_r(s_R)$, Eq. (14)
$\mathcal{U}_5$	drilling rotation	not a degree of freedom of the plate theory

Inserting Eqs. (1) and (2) into Eq. (10) gives, e.g. for $\mathcal{U}_3$,

$$\left[u_k(s_L)-u_k(s_R)\right]\phi_{k,x}(x)+\left[u_j(s_L)-u_j(s_R)\right]\varphi_j(x)=0\qquad\forall\,x\in[0,L]. \tag{11}$$

Proposition 1. Equation (10) holds for arbitrary amplitude functions if, and only if, every mode satisfies it individually: $\mathbf{U}_k$, θ_k and u_k of all conventional modes, u_j of all shear modes and ξ_r, ζ_r of all Mindlin modes take equal values at s_L and s_R.

Proof. If every bracket vanishes, Eq. (11) and its analogues hold for any amplitudes. Conversely, if all amplitude functions but one are set to zero, the bracket of the remaining mode must vanish. □

The periodic condition can therefore be built into the cross-section analysis by modes that are periodic one by one (Sections 2.2.2–2.2.4); every combination of them, in particular every Ritz trial function, is then periodic, without penalty terms or Lagrange multipliers.

2.2.2. Conventional modes: constrained family subspaces

For the conventional modes, the conditions on $\mathcal{U}_1$, $\mathcal{U}_2$ and the nodal rotation part of $\mathcal{U}_6$ are three linear equations,

$$\mathbf{R}\begin{bmatrix}\mathbf{U}\\ \Theta\end{bmatrix}=\mathbf{0},\qquad \mathbf{R}=\begin{bmatrix}\mathbf{e}_{X,s_L}^{\mathsf{T}}-\mathbf{e}_{X,s_R}^{\mathsf{T}}\\ \mathbf{e}_{Y,s_L}^{\mathsf{T}}-\mathbf{e}_{Y,s_R}^{\mathsf{T}}\\ \mathbf{e}_{\theta,s_L}^{\mathsf{T}}-\mathbf{e}_{\theta,s_R}^{\mathsf{T}}\end{bmatrix}\in\mathbb{R}^{3\times 3N},\qquad \operatorname{rank}\mathbf{R}=3, \tag{12}$$

where the unit vectors $\mathbf{e}$ pick the U_X, U_Y or θ entry of a side node and the three rows are independent ($\operatorname{rank}\mathbf{R}=3$). The periodic deformations form the null space $\ker\mathbf{R}$ of dimension $3N-3$. To keep the GBT meaning of the modes, they are built family by family. With $\mathbf{V}_g$ the nodal fields of family $g\in\{G,D,L,T\}$ of the non-periodic cell: (1) columns with $\mathbf{R}\mathbf{v}=\mathbf{0}$ are kept; (2) the other columns $\mathbf{V}_g'$ are replaced by the largest periodic subspace $\mathbf{V}_g'\mathbf{Z}_g$, with $\mathbf{Z}_g$ a basis of $\ker(\mathbf{R}\mathbf{V}_g')$, i.e. of all $\mathbf{z}$ with $\mathbf{R}\mathbf{V}_g'\mathbf{z}=\mathbf{0}$; (3) periodic deformations not reached by steps 1–2, $(\mathbf{I}-\mathbf{P})\ker\mathbf{R}$ with $\mathbf{P}$ the orthogonal projector onto their span, are added to T if they elongate a segment and to D otherwise; (4) every mode is scaled to a unit maximum. The columns of the modal matrix $\mathbf{V}$ then span exactly $\ker\mathbf{R}$: no admissible deformation is lost and none is spurious ($n_C=3N-3$). Family G keeps only the two translations, since a rigid rotation ω about x would lift one side edge and lower the other ($U_{Y,s_L}-U_{Y,s_R}=-W\omega$). Equal lateral translations and rotations of the side nodes enter family D in step 3.

2.2.3. Warping: the periodic cell as a closed contour

The condition $u_k(s_L) = u_k(s_R)$ joins the side edges, so that for the warping the cell is a contour closed through its side edges, like the closed sections of GBT [24,40]. With the segment difference matrix Δ, $(\Delta\mathbf{u})_p = u_{q'} - u_q$, the weights $\mathbf{A}_w = \mathrm{diag}(\hat{A}_{66,p}/b_p)$ and the vector $\mathbf{f}$ of $b_p v_p(\eta = 1/2)$, the objective of Eq. (9) is $(\Delta\mathbf{u}+\mathbf{f})^\mathsf{T}\mathbf{A}_w(\Delta\mathbf{u}+\mathbf{f})$; with the constraints $\mathbf{C}_u\mathbf{u} = \mathbf{0}$, $\mathbf{C}_u = [\mathbf{c}_{ax}^\mathsf{T};\ \mathbf{e}_{s_L}^\mathsf{T} - \mathbf{e}_{s_R}^\mathsf{T}]$, its minimum is the solution of the standard Karush–Kuhn–Tucker system

$$\begin{bmatrix} \Delta^\mathsf{T}\mathbf{A}_w\Delta & \mathbf{C}_u^\mathsf{T} \\ \mathbf{C}_u & \mathbf{0} \end{bmatrix} \begin{bmatrix} \mathbf{u} \\ \tau \end{bmatrix} = \begin{bmatrix} -\Delta^\mathsf{T}\mathbf{A}_w\mathbf{f} \\ \mathbf{0} \end{bmatrix}, \tag{13}$$

with the Lagrange multipliers τ. The solution is unique, since the axial constraint excludes uniform warping, the only warping without shear energy. For a mode with non-zero circulation $\oint v\,ds$, the remaining γ_k is the shear flow of a closed section, carried by $A_{66}\gamma_k\gamma_i$ in the modal tensors.

2.2.4. Shear and Mindlin modes: merging of the side-edge functions

The shear and Mindlin modes are defined node by node (hat function $\hat{u}_q$, vectors Ω_q and function ζ_q of node q). A combination with the coefficients y_q is periodic if, and only if, $y_{s_L} = y_{s_R}$, so the periodic basis replaces the separate functions of the two side nodes by their sums,

$$\bar{u} = \hat{u}_{s_L} + \hat{u}_{s_R}, \qquad \bar{\Omega} = \Omega_{s_L} + \Omega_{s_R}, \qquad \bar{\zeta} = \zeta_{s_L} + \zeta_{s_R}, \tag{14}$$

while the functions of all other nodes are kept. This basis spans exactly the periodic subspace (Ω_{s_L} and Ω_{s_R} can be paired because both side nodes lie in the straight skin). This gives $n_S = N-1$ shear modes, including the axial mode, and $n_M = \sum_q m_q - m_{s_L} + N - 1$ Mindlin modes, with m_q independent vectors Ω_q at node q ($m_q = 2$ at a kink, $m_q = 1$ between collinear segments); see Table 2.

Table 2. Number of deformation modes per family of the periodic cell (independent of the lay-up).

b_s [mm]	N	G	D	L	T	S (incl. axial)	M	total
150	92	2	13	166	92	91	194	558
285.5	78	2	13	138	78	77	166	474

2.2.5. Properties and hierarchical ordering

The periodic basis is exact (every truncation satisfies Eq. (10)), complete (the full basis is equivalent to imposing Eq. (10) node by node), interpretable through the GBT families and consistent with Bloch–Floquet theory [8]: a cell of width $2b_s$ contains the Bloch waves $\mathbf{u}(X + b_s, x) = e^{\mathrm{i}k_B b_s}\mathbf{u}(X, x)$ with the Bloch wavenumbers $k_B b_s = 0$ and π, i.e. modes that repeat in every bay and modes that alternate. To order them from global to local, the conventional modes are recombined by $\hat{\mathbf{B}}\,\mathbf{y} = \Lambda(k_r^2\hat{\mathbf{D}}^I + k_r^4\hat{\mathbf{C}})\,\mathbf{y}$ (matrices of the modal tensors, Eq. (25); reference wavenumber k_r); a small ratio Λ of transverse to longitudinal stiffness indicates a global mode, and the recombined modes remain periodic. In Figure 2, modes 1 and 2 are rigid translations; the others combine skin-bay bending with stringer translation, rotation and distortion and, unlike modes of open members, belong to no single family.

2.3. Member analysis

2.3.1. Strain energy and modal tensors

To obtain the equations for the amplitude functions, the strain energy is expressed in terms of the modes. Its variation

$$\delta U = \int_L \int_b \int_t \sigma^\mathsf{T}\delta\varepsilon\, dz\, ds\, dx = \int_L \int_b \left(\mathbf{N}^\mathsf{T}\delta\varepsilon^M + \mathbf{M}^\mathsf{T}\delta\kappa + \mathbf{Q}^\mathsf{T}\delta\gamma_z\right) ds\, dx, \tag{15}$$

(integration over the length L, contour b and thickness t) becomes, with Eqs. (3) and (5) and integration over s, in the form of Silvestre and Camotim [32],

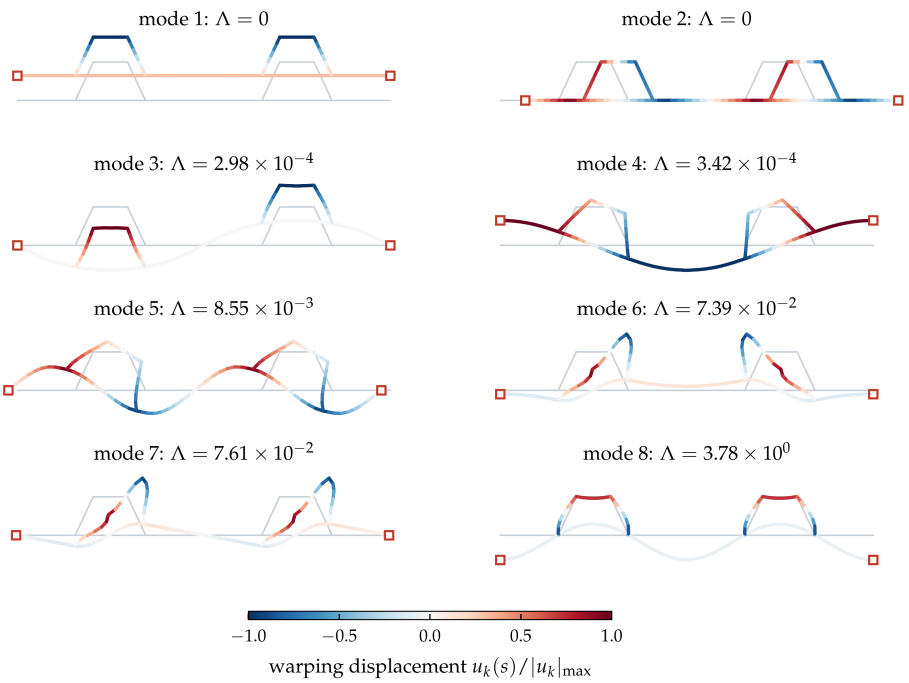


Figure 2. First eight hierarchical conventional modes of the periodic cell ($b_s = 150$ mm, Cross-ply lay-up, ordered at $k_r = \pi/b_s$). Line: in-plane deformation (grey: undeformed); colour: normalised warping $u_k(s)$. The side nodes s_L and s_R (squares) move identically in every mode.

$$\begin{aligned}\delta U = \int_L \Big(& C_{ik}\phi_{k,xx}\delta\phi_{i,xx} + H^I_{ik}\phi_{k,x}\delta\phi_{i,xx} + H^I_{ki}\phi_{k,xx}\delta\phi_{i,x} + D^I_{ik}\phi_{k,x}\delta\phi_{i,x} \\ & + D^{II}_{ik}\phi_k\delta\phi_{i,xx} + D^{II}_{ki}\phi_{k,xx}\delta\phi_i + F^I_{ik}\phi_{k,x}\delta\phi_i + F^I_{ki}\phi_k\delta\phi_{i,x} + B_{ik}\phi_k\delta\phi_i \\ & + C_{hj}\varphi_{j,x}\delta\varphi_{h,x} + H^I_{hj}\varphi_j\delta\varphi_{h,x} + H^I_{jh}\varphi_{j,x}\delta\varphi_h + D^I_{hj}\varphi_j\delta\varphi_h \\ & + C_{ij}\varphi_{j,x}\delta\phi_{i,xx} + H^I_{ij}\varphi_j\delta\phi_{i,xx} + H^{II}_{ji}\varphi_{j,x}\delta\phi_{i,x} \\ & + D^I_{ji}\varphi_j\delta\phi_{i,x} + D^{II}_{ji}\varphi_{j,x}\delta\phi_i + F^{II}_{ij}\varphi_j\delta\phi_i \\ & + C_{hk}\phi_{k,xx}\delta\varphi_{h,x} + H^{II}_{kh}\phi_{k,xx}\delta\varphi_h + H^I_{hk}\phi_{k,x}\delta\varphi_{h,x} \\ & + D^I_{hk}\phi_{k,x}\delta\varphi_h + D^{II}_{hk}\phi_k\delta\varphi_{h,x} + F^{II}_{kh}\phi_k\delta\varphi_h \Big) dx + \delta U_M,\end{aligned} \tag{16}$$

where the first two lines contain the conventional modes, the third the shear modes, the others their coupling, and δU_M the Mindlin terms. The tensors of the last two lines follow from those of the preceding two by exchanging conventional and shear modes ($C_{ij} \to C_{hk}$, $H^I_{ij} \to H^{II}_{kh}$, $H^{II}_{ji} \to H^I_{hk}$, $D^I_{ji} \to D^I_{hk}$, $D^{II}_{ji} \to D^{II}_{hk}$, $F^{II}_{ij} \to F^{II}_{kh}$, as in Eq. (12) of [32]). C pairs axial, B transverse, D^I shear, D^{II} axial–transverse, F transverse–shear and H axial–shear strains. For the conventional modes,

$$C_{ik} = \int_b \Big[A_{11}\, u_k u_i - B_{11}\,(w_k u_i + w_i u_k) + D_{11}\, w_k w_i \Big] ds, \tag{17}$$

$$B_{ik} = \int_b \Big[A_{22}\, v_{k,s} v_{i,s} - B_{22}\,(v_{k,s} w_{i,ss} + w_{k,ss} v_{i,s}) + D_{22}\, w_{k,ss} w_{i,ss} \Big] ds, \tag{18}$$

$$D^I_{ik} = \int_b \Big[A_{66}\, \gamma_k \gamma_i - 2B_{66}\,(\gamma_k w_{i,s} + w_{k,s}\gamma_i) + 4D_{66}\, w_{k,s} w_{i,s} \Big] ds, \tag{19}$$

$$D^{II}_{ik} = \int_b \Big[A_{12}\, v_{k,s} u_i - B_{12}\,(v_{k,s} w_i + w_{k,ss} u_i) + D_{12}\, w_{k,ss} w_i \Big] ds, \tag{20}$$

$$F^I_{ik} = \int_b \Big[A_{26}\,\gamma_k v_{i,s} - B_{26}\left(\gamma_k w_{i,ss} + 2 w_{k,s} v_{i,s}\right) + 2D_{26}\, w_{k,s} w_{i,ss} \Big]\, ds, \tag{21}$$

$$H^I_{ik} = \int_b \Big[A_{16}\,\gamma_k u_i - B_{16}\left(\gamma_k w_i + 2 w_{k,s} u_i\right) + 2D_{16}\, w_{k,s} w_i \Big]\, ds; \tag{22}$$

for the shear modes,

$$C_{hj} = \int_b A_{11}\, u_j u_h\, ds, \qquad D^I_{hj} = \int_b A_{66}\, u_{j,s} u_{h,s}\, ds, \qquad H^I_{hj} = \int_b A_{16}\, u_{j,s} u_h\, ds; \tag{23}$$

and for the coupling between conventional and shear modes,

$$\begin{aligned} C_{ij} &= \int_b (A_{11} u_i - B_{11} w_i)\, u_j\, ds, & H^I_{ij} &= \int_b (A_{16} u_i - B_{16} w_i)\, u_{j,s}\, ds, \\ H^{II}_{ji} &= \int_b \left(A_{16}\gamma_i - 2B_{16} w_{i,s}\right) u_j\, ds, & D^I_{ji} &= \int_b \left(A_{66}\gamma_i - 2B_{66} w_{i,s}\right) u_{j,s}\, ds, \\ D^{II}_{ji} &= \int_b \left(A_{12} v_{i,s} - B_{12} w_{i,ss}\right) u_j\, ds, & F^{II}_{ij} &= \int_b \left(A_{26} v_{i,s} - B_{26} w_{i,ss}\right) u_{j,s}\, ds. \end{aligned} \tag{24}$$

For modes satisfying Vlasov's assumptions, Eqs. (17)–(24) reduce to Eqs. (10)–(12) of [32]; the terms with γ_k and $v_{k,s}$ are the additions for closed contours and transverse extension. In the block-matrix notation of the GBT–Ritz approach [50], the conventional tensors form the matrices $\hat{\mathbf{B}} = [B_{ik}]$, $\hat{\mathbf{C}} = [C_{ik}]$, etc. (the hat distinguishes them from the laminate stiffnesses **A**, **B**, **D** of Eq. (5)) and the block matrix

$$\Xi = \begin{bmatrix} \hat{\mathbf{B}} & \hat{\mathbf{F}}^I & (\hat{\mathbf{D}}^{II})^{\mathsf{T}} \\ (\hat{\mathbf{F}}^I)^{\mathsf{T}} & \hat{\mathbf{D}}^I & (\hat{\mathbf{H}}^I)^{\mathsf{T}} \\ \hat{\mathbf{D}}^{II} & \hat{\mathbf{H}}^I & \hat{\mathbf{C}} \end{bmatrix}, \qquad \delta U = \int_L \delta\Phi^{\mathsf{T}} \Xi\, \Phi\, dx, \qquad \Phi = \begin{bmatrix} \phi \\ \phi_{,x} \\ \phi_{,xx} \end{bmatrix}, \tag{25}$$

where ϕ collects the amplitudes ϕ_k (the blocks $\hat{\mathbf{D}}_2$ and $\hat{\mathbf{H}}_1$ of [50] are $(\hat{\mathbf{D}}^{II})^{\mathsf{T}}$ and $(\hat{\mathbf{H}}^I)^{\mathsf{T}}$ here). For all modes, including δU_M, writing the strains of Eq. (5) as $[\varepsilon^M; \kappa; \gamma_z] = \sum_{d=0}^{2} \mathcal{E}^{(d)}(s)\, \Phi^{(d)}(x)$, where $\Phi^{(d)}$ collects the d-th derivatives of all amplitude functions ($\Phi = [\Phi^{(0)}; \Phi^{(1)}; \Phi^{(2)}]$) and $\mathcal{E}^{(d)}$ contains the corresponding strain profiles of Eq. (3), turns Eq. (15) into

$$\delta U = \int_L \sum_{d,e=0}^{2} \delta\Phi^{(d)\mathsf{T}}\, \Xi^{(de)}\, \Phi^{(e)}\, dx, \qquad \Xi^{(de)} = \int_b \mathcal{E}^{(d)\mathsf{T}}\, \mathbf{C}_{\text{lam}}\, \mathcal{E}^{(e)}\, ds, \tag{26}$$

with $\mathbf{C}_{\text{lam}}$ the laminate stiffness matrix of Eq. (5). For a Mindlin mode r, Eq. (3) gives the non-zero entries $\zeta_{r,s}\,(\kappa_{ss})$, $\xi_{r,s}\,(\kappa_{xs})$, $\xi_r\,(\gamma_{xz})$, $\zeta_r\,(\gamma_{sz})$ of $\mathcal{E}^{(0)}$ and $\xi_r\,(\kappa_{xx})$, $\zeta_r\,(\kappa_{xs})$ of $\mathcal{E}^{(1)}$; for the conventional and shear modes, Eq. (26) reproduces Eqs. (17)–(24).

2.3.2. Geometric tensors, loads and pre-buckling state

Buckling occurs when the work of the pre-buckling membrane forces S_{xx}, S_{ss}, S_{xs} (per unit length, tension positive) on the non-linear strains of Eq. (4) balances the strain energy of the mode. This second-order work

$$V = \int_L \int_b (S_{xx}\, \varepsilon^{NL}_{xx} + S_{ss}\, \varepsilon^{NL}_{ss} + S_{xs}\, \gamma^{NL}_{xs})\, ds\, dx, \tag{27}$$

defines the geometric tensors; with Eq. (1), for the conventional modes,

$$
\begin{aligned}
\delta V = \int_L \Big(& X^U_{ik}\,\phi_{k,xx}\delta\phi_{i,xx} + X^{VW}_{ik}\,\phi_{k,x}\delta\phi_{i,x} + X^S_{ik}\,\phi_k\delta\phi_i \\
& + X^P_{ik}\,\phi_{k,x}\delta\phi_i + X^P_{ki}\,\phi_k\delta\phi_{i,x} + X^R_{ik}\,\phi_{k,xx}\delta\phi_{i,x} + X^R_{ki}\,\phi_{k,x}\delta\phi_{i,xx}\Big)dx + \dots , \\
X^U_{ik} = \int_b S_{xx}\,u_i u_k\,ds, \qquad & X^{VW}_{ik} = \int_b \left[S_{xx}\,(v_i v_k + w_i w_k) + S_{ss}\,u_{i,s}u_{k,s}\right] ds, \\
X^S_{ik} = \int_b S_{ss}\,(v_{i,s}v_{k,s} + w_{i,s}w_{k,s})\,ds, \qquad & X^P_{ik} = \int_b S_{xs}\,(v_{i,s}v_k + w_{i,s}w_k)\,ds, \\
& X^R_{ik} = \int_b S_{xs}\,u_{i,s}u_k\,ds,
\end{aligned}
\tag{28}
$$

where the dots stand for the shear-mode terms. In the general form analogous to Eq. (26), with the displacement gradients of all modes $\partial[u,v,w]^{\mathsf T}/\partial x = \sum_d \mathbf{J}^{(d)}_x \Phi^{(d)}$ and $\partial[u,v,w]^{\mathsf T}/\partial s = \sum_d \mathbf{J}^{(d)}_s \Phi^{(d)}$,

$$
\mathcal{X}^{(de)} = \int_b \Big[S_{xx}\,\mathbf{J}^{(d)\mathsf T}_x \mathbf{J}^{(e)}_x + S_{ss}\,\mathbf{J}^{(d)\mathsf T}_s \mathbf{J}^{(e)}_s + S_{xs}\left(\mathbf{J}^{(d)\mathsf T}_x \mathbf{J}^{(e)}_s + \mathbf{J}^{(d)\mathsf T}_s \mathbf{J}^{(e)}_x\right)\Big] ds,
\tag{29}
$$

which contains Eq. (28) and, with $\partial u/\partial x = u_j\varphi_{j,x}$ and $\partial u/\partial s = u_{j,s}\varphi_j$, the shear-mode terms; the Mindlin modes do not move the mid-surface and do not contribute. For a uniform S_{xx} and $S_{ss} = S_{xs} = 0$, only X^U_{ik} and X^{VW}_{ik} remain, the geometric matrices of the GBT–Ritz approach [47,50]; X^{VW}_{ik} corresponds to $X_{pik}W^o_p$ of [31,32] (there, p is a mode index).

The compressive end load $n_0(s)$, acting at the distance $e_0(s)$ from the wall reference surface, enters through generalised end forces, as q^I_i in [32]; with the axial displacement $u_0 = u - e_0 w_{,x}$ at the load,

$$
\begin{aligned}
\delta\Pi_{ext} = & -\left(q^I_k\,\delta\phi_{k,x} + q^I_h\,\delta\varphi_h\right)_{x=0} + \left(q^I_k\,\delta\phi_{k,x} + q^I_h\,\delta\varphi_h\right)_{x=L}, \\
q^I_k = \int_b n_0\,(u_k - e_0 w_k)\,ds, \qquad & q^I_h = \int_b n_0\,u_h\,ds.
\end{aligned}
\tag{30}
$$

Instead of assuming stresses through modal resultants W^o_p [32], the pre-buckling state is computed, since the load enters through the eccentric foot, is shared by shear lag and produces transverse forces (Section 3.2). It is the linear static solution of the same GBT–Ritz model under the end load, $\delta U + \delta\Pi_{ext} = 0$, which gives the membrane strains $\varepsilon^{M,0}$, the curvatures κ^0 and the forces

$$
[S_{xx},\ S_{ss},\ S_{xs}]^{\mathsf T} = \mathbf{A}\,\varepsilon^{M,0} + \mathbf{B}\,\kappa^0,
\tag{31}
$$

which vary along s and x.

2.3.3. Equilibrium equations and boundary conditions

At the critical load, $\delta U + \lambda\,\delta V = 0$ for all variations, with the load factor λ scaling the pre-buckling forces. Integration by parts in x gives one GBT equilibrium equation per conventional and per shear mode (the Mindlin equations follow likewise),

$$
\begin{aligned}
\mathcal{K}_{ik}\,\phi_k + \mathcal{K}_{ij}\,\varphi_j + \lambda\left(\mathcal{G}_{ik}\,\phi_k + \mathcal{G}_{ij}\,\varphi_j\right) = 0, \\
\mathcal{K}_{hk}\,\phi_k + \mathcal{K}_{hj}\,\varphi_j + \lambda\left(\mathcal{G}_{hk}\,\phi_k + \mathcal{G}_{hj}\,\varphi_j\right) = 0,
\end{aligned}
\tag{32}
$$

with the stiffness operators

$$\begin{aligned}
\mathcal{K}_{ik} &= C_{ik}\frac{d^4}{dx^4} + H_{ik}\frac{d^3}{dx^3} - D_{ik}\frac{d^2}{dx^2} + F_{ik}\frac{d}{dx} + B_{ik},\\
\mathcal{K}_{ij} &= C_{ij}\frac{d^3}{dx^3} + \left(H^I_{ij} - H^{II}_{ji}\right)\frac{d^2}{dx^2} - \left(D^I_{ji} - D^{II}_{ji}\right)\frac{d}{dx} + F^{II}_{ij},\\
\mathcal{K}_{hk} &= C_{hk}\frac{d^3}{dx^3} + \left(H^I_{hk} - H^{II}_{kh}\right)\frac{d^2}{dx^2} - \left(D^I_{hk} - D^{II}_{hk}\right)\frac{d}{dx} - F^{II}_{kh},\\
\mathcal{K}_{hj} &= C_{hj}\frac{d^2}{dx^2} + H_{hj}\frac{d}{dx} - D^I_{hj},
\end{aligned} \tag{33}$$

where the shear-mode equation is multiplied by -1, as $\mathcal{K}_{hj}$ in [32], and

$$H_{ik} = H^I_{ik} - H^I_{ki}, \qquad D_{ik} = D^I_{ik} - D^{II}_{ik} - D^{II}_{ki}, \qquad F_{ik} = F^I_{ik} - F^I_{ki}, \qquad H_{hj} = H^I_{hj} - H^I_{jh}. \tag{34}$$

The geometric operators $\mathcal{G}$ follow from Eq. (28); for a uniform S_{xx}, $\mathcal{G}_{ik} = X^U_{ik}\,d^4/dx^4 - X^{VW}_{ik}\,d^2/dx^2$, in general with x-dependent coefficients. The boundary terms give the end conditions

$$\left[\left(W^\sigma_i + q^I_i\right)\delta\phi_{i,x} + W^\tau_i\,\delta\phi_i + \left(W^\sigma_h + q^I_h\right)\delta\varphi_h\right]^{x=L}_{x=0} = 0, \tag{35}$$

with the generalised normal stress resultants

$$\begin{aligned}
W^\sigma_i &= C_{ik}\phi_{k,xx} + H^I_{ik}\phi_{k,x} + D^{II}_{ik}\phi_k + C_{ij}\varphi_{j,x} + H^I_{ij}\varphi_j,\\
W^\sigma_h &= C_{hj}\varphi_{j,x} + H^I_{hj}\varphi_j + C_{hk}\phi_{k,xx} + H^I_{hk}\phi_{k,x} + D^{II}_{hk}\phi_k,
\end{aligned} \tag{36}$$

and the generalised shear stress resultant $W^\tau_i = -W^\sigma_{i,x} + W^{xs}_i$, $W^{xs}_i = H^I_{ki}\phi_{k,xx} + D^I_{ik}\phi_{k,x} + F^I_{ki}\phi_k + H^{II}_{ji}\varphi_{j,x} + D^I_{ji}\varphi_j$. Since the coefficients vary with x and the end conditions couple all modes, these equations are solved by the Ritz method, directly from the energy.

2.3.4. Ritz solution, end conditions and classical limit

As in [47,50], every amplitude function is a sum of basis functions with unknown coefficients,

$$\phi_k(x) = \sum_{l=1}^{n_b}\Psi_l(x)\,a_{kl}, \qquad \varphi_j(x) = \sum_{l=1}^{n_b}\Psi_l(x)\,a_{jl}, \qquad \chi_r(x) = \sum_{l=1}^{n_b}\Psi_l(x)\,a_{rl}, \tag{37}$$

i.e. $\Phi = \Psi\mathbf{a}$, with Ψ the matrix of the basis functions Ψ_l and their derivatives and $\mathbf{a}$ all coefficients. Instead of trigonometric series [47,50], the n_b functions Ψ_l are cubic B-splines on an open uniform knot vector with spans $h_x = b_s/20$: they are local, so that they resolve the end zones, and their end values are the end coefficients ($\phi(0) = a_1$; $\phi_{,x}(0) = 0 \Leftrightarrow a_2 = a_1$), so that end conditions are imposed exactly. Cubic B-splines are established in spline finite strips [12] and isogeometric analysis [55]; unlike the Hermite elements of GBT-based finite elements [38], they are twice continuously differentiable.

As in the shell models (Section 3), **simply supported (SS)** loaded edges have $U_Y = 0$ at every end node, i.e. $\sum_k U_{Y,qk}\phi_k = 0$ at $x = 0, L$, imposed by reducing $\mathbf{a}$ to the null space of these conditions; unlike the classical $\phi_k = \sin(m\pi x/L)$, this leaves U_X, warping and rotations free. **Clamped (CC)** edges have $\phi_k = \phi_{k,x} = 0$ and $\chi_r = 0$ at both ends and $\varphi_j(0) = 0$, so that $x = L$ can warp under the load. Inserting $\Phi = \Psi\mathbf{a}$ into Eqs. (26) and (29) turns $\delta U + \lambda\,\delta V = 0$ into

$$\left(\mathbf{K} + \lambda\,\mathbf{G}\right)\mathbf{a} = \mathbf{0}, \qquad \mathbf{K} = \int_0^L \Psi^{\mathsf{T}}\Xi\,\Psi\,dx, \qquad \mathbf{G} = \int_0^L \Psi^{\mathsf{T}}\mathcal{X}(x)\,\Psi\,dx, \tag{38}$$

with $\Xi = [\Xi^{(de)}]$ and $\mathcal{X} = [\mathcal{X}^{(de)}]$ (four Gauss points per span). With pre-buckling forces for $n_0 = 1$ N/mm, the smallest positive λ is N_{cr} in N/mm. Since $\mathbf{G}$ is indefinite, the largest $\mu = 1/\lambda$ of $-\mathbf{G}\mathbf{a} = \mu\mathbf{K}\mathbf{a}$

is computed by the implicitly restarted Lanczos method [56]. For $b_s = 150$ mm and $L = 225$ mm (33 B-splines), the SS model has 18,230 unknowns.

Classical GBT is a special case, used for verification (Section 3.1): for uncoupled walls ($A_{16} = A_{26} = 0$, $\mathbf{B} = \mathbf{0}$, $D_{16} = D_{26} = 0$), a uniform S_{xx} and conventional modes only, $H_{ik} = F_{ik} = 0$ and Eq. (32) becomes

$$C_{ik}\,\phi_{k,xxxx} - D_{ik}\,\phi_{k,xx} + B_{ik}\,\phi_k + \lambda\left(X^U_{ik}\,\phi_{k,xxxx} - X^{VW}_{ik}\,\phi_{k,xx}\right) = 0, \tag{39}$$

which, without X^U_{ik}, is the classical second-order GBT equation [24,31]; with $\phi_k = \hat{\phi}_k \sin(k_m x)$, $k_m = m\pi/L$, for m half-waves,

$$\left[B_{ik} + k_m^2\,D_{ik} + k_m^4\,C_{ik} + \lambda\left(k_m^2\,X^{VW}_{ik} + k_m^4\,X^U_{ik}\right)\right]\hat{\phi}_k = 0, \tag{40}$$

whose smallest eigenvalue over m is the signature curve. The periodic condition does not change this structure; it restricts the admissible modes.

2.4. Mode decomposition

As in [50], the buckling mode is decomposed into the contributions of the families $g \in \{G, D, L, T, S, M\}$: with $\mathbf{a}_g$ the vector $\mathbf{a}$ with all coefficients outside family g set to zero, the strain energy $U = \frac{1}{2}\mathbf{a}^\mathsf{T}\mathbf{K}\mathbf{a}$ is divided into

$$U_g = \tfrac{1}{2}\,\mathbf{a}_g^\mathsf{T}\mathbf{K}\,\mathbf{a}, \qquad \rho_g = \frac{U_g}{U}, \qquad \sum_g \rho_g = 1. \tag{41}$$

The participation ρ_g includes half of the coupling energy between g and every other family, so that the participations add up exactly to one; the normalised self-energies $\frac{1}{2}\mathbf{a}_g^\mathsf{T}\mathbf{K}\mathbf{a}_g$ of [50] omit the coupling energy. This matters for the strongly coupled D and L modes: for the Unsym cell ($b_s = 150$ mm, $\beta = 1.5$, SS) the normalised self-energies assign 41 % to D and 58 % to L, Eq. (41) 1 % and 97 %. Since $\mathbf{K}\mathbf{a} = -\lambda\mathbf{G}\mathbf{a}$ and the Mindlin modes do not enter $\mathbf{G}$, $\rho_M = 0$; their influence is assessed by omitting them (Section 3.6). The partition also applies to individual hierarchical modes.

3. Results and discussion

After a verification (Section 3.1) and the pre-buckling state (Section 3.2), critical loads and modes are compared with Abaqus shell models (Sections 3.3–3.5), followed by truncated bases (Section 3.6) and accuracy, cost and limitations (Section 3.7). Tables 3 and 4 give the geometry, materials and lay-ups: an isotropic reference, an uncoupled cross-ply laminate, two angle-ply laminates with bending–twisting coupling and an unsymmetric laminate with membrane–bending coupling. The pitches $b_s = 150$ mm and 285.5 mm with $0.5 \le \beta \le 3$ are considered first; longer cells follow in Section 3.5.

Table 3. Geometry of the cross-section (Figure 1) and material properties (CFRP: carbon fibre-reinforced polymer).

Parameter	Value
stringer pitch b_s	150 mm, 285.5 mm
web root half-width b_3, web height h_w, web inclination	27.75 mm, 30 mm, 25° from the vertical
crown half-width $b_t = b_3 - h_w \tan 25°$	13.76 mm
foot width $b_1 = 0.1\,(b_s - 2b_3)$	9.45 mm, 23.00 mm
skin and stringer thickness t_s, t_f	1.0 mm, 0.875 mm (8 and 7 plies of 0.125 mm)
stringer offset $c = (t_s + t_f)/2$	0.9375 mm
CFRP ply	$E_1 = 150$ GPa, $E_2 = 9$ GPa, $G_{12} = G_{13} = G_{23} = 5.3$ GPa, $\nu_{12} = 0.33$
isotropic material	$E = 200$ GPa, $\nu = 0.3$

Table 4. Lay-ups of skin and stringer (angles in degrees from the x-axis).

Name	Skin	Stringer	Coupling
Isotropic	–	–	none
Cross-ply	$[0/90/0/90]_s$	$[0/90/0/90/0/90/0]$	none
Sym1	$[45/-45/-45/45]_s$	$[45/-45/45/-45/45/-45/45]$	D_{16}, D_{26} (stringer also A_{16}, A_{26})
Sym2	$[60/-30/-30/60]_s$	$[60/-30/60/-30/60/-30/60]$	$A_{16}, A_{26}, D_{16}, D_{26}$
Unsym	$[0/30/60/90]_2$	$[0/30/60/90/0/30/60]$	**A**, **B**, **D** fully populated

In both models, a line load of 1 N/mm compresses both loaded edges of skin, webs and crowns, and 2 N/mm a foot (two edges: skin and flange) at the height of their resultant, so that λ is directly N_{cr} in N/mm. The shell models are those of [17,22] (geometry, S4R/S3 elements with a seed of 3.7 mm, stringer tied to the skin by *TIE with the true offset c), extended here by periodic constraint equations; they are linear buckling analyses in Abaqus/Standard 2024. Eq. (10) links every skin node of the edge $X = -W$ to the node of $X = 0$ at the same x by one constraint equation per degree of freedom, except for degrees of freedom already prescribed at the end corners. Simply supported (SS) loaded edges have $\mathcal{U}_2 = 0$ on every end node, with $\mathcal{U}_1 = \mathcal{U}_3 = 0$ at one corner; clamped (CC) loaded edges have all six degrees of freedom fixed at $x = 0$ and all except $\mathcal{U}_3$ at $x = L$, where the load is applied. GBT imposes the same conditions (Section 2.3.4). The GBT–Ritz analysis was implemented in Python 3.13 (NumPy 2.5, SciPy 1.18). The deviation of GBT from Abaqus is $e = (N_{cr}^{\text{GBT}} - N_{cr}^{\text{Abaqus}})/N_{cr}^{\text{Abaqus}}$, and modes are compared by the modal assurance criterion $\text{MAC} = (\mathbf{w}_{\text{GBT}}^{\mathsf{T}}\mathbf{w}_{\text{Abq}})^2/[(\mathbf{w}_{\text{GBT}}^{\mathsf{T}}\mathbf{w}_{\text{GBT}})(\mathbf{w}_{\text{Abq}}^{\mathsf{T}}\mathbf{w}_{\text{Abq}})]$ of the skin deflections at all skin nodes of the shell model ($\mathbf{w}_{\text{Abq}}$) and of GBT interpolated to them ($\mathbf{w}_{\text{GBT}}$); MAC = 1 means identical shapes.

3.1. Verification

Three checks confirm the implementation. Every mode is periodic to round-off: for the cross-ply cell with $b_s = 150$ mm, the largest side-node differences of the conventional modes are 7.9×10^{-15} (in-plane displacements), 1.1×10^{-16} (rotation) and 2.3×10^{-11} (warping), and zero for the shear and Mindlin modes. The explicit tensors of Eqs. (17)–(24) agree with the numerical cross-section integration of Eq. (26) to 2×10^{-15}, and, in the classical limit (Section 2.3.4), the Ritz solution reproduces the signature-curve minimum of Eq. (40) for the cross-ply cell (33.747 N/mm, three half-waves for $L = 225$ mm and six for $L = 450$ mm) to 0.001 %.

3.2. Pre-buckling state

Figure 3 shows the computed membrane forces for $b_s = 150$ mm and $\beta = 1.5$. Away from the loaded edges the axial force is 1.00–$1.16\, n_0$ in the skin and 1.6–$2.0\, n_0$ in the feet (Figure 3a,c), and the transverse force (Figure 3b) is nearly uniform: $0.32\, n_0$ for the isotropic, 0.03–$0.04\, n_0$ for the Cross-ply and Unsym, and 0.86–$0.89\, n_0$ for the angle-ply cells. It results from the restrained Poisson expansion: the compressed skin tends to expand laterally, but the periodic condition, like the shell model's constraint equations, keeps the cell width constant, so that $S_{ss} \approx (A_{12}/A_{11})\, S_{xx}$; for the isotropic, Cross-ply and Sym1 skins the ratio agrees within 2 %, and the angle-ply values are large because of their large A_{12}. This compression enters **G** through Eq. (28) and becomes decisive for long cells (Section 3.5). Clamping raises the axial force at $x = 0$, up to $1.43\, n_0$ in the skin bay of the Sym1 cell (Figure 3c,d).

3.3. Critical loads of short cells

Figure 4 and Table 5 give the critical loads of the 60 cases, and Figure 5 the deviations from Abaqus. With SS loaded edges, GBT agrees with Abaqus within 0.57 % on average and 1.25 % at most, with the same accuracy for uncoupled and coupled laminates (mean 0.49–0.73 % for every lay-up). With CC loaded edges the deviations are 0.64 % on average and 1.52 % at most. The SS curves of the isotropic, cross-ply and unsymmetric cells vary little with β, with the garland maxima of plates where the number of half-waves changes (e.g. $\beta = 0.75$, cross-ply, $b_s = 150$ mm); the angle-ply curves decrease monotonically, because skewed buckles cannot adapt to straight loaded edges. Clamping raises the critical loads of short cells, by 59 % for the isotropic cell with $b_s = 150$ mm and $\beta = 0.5$, and all CC curves decrease monotonically.

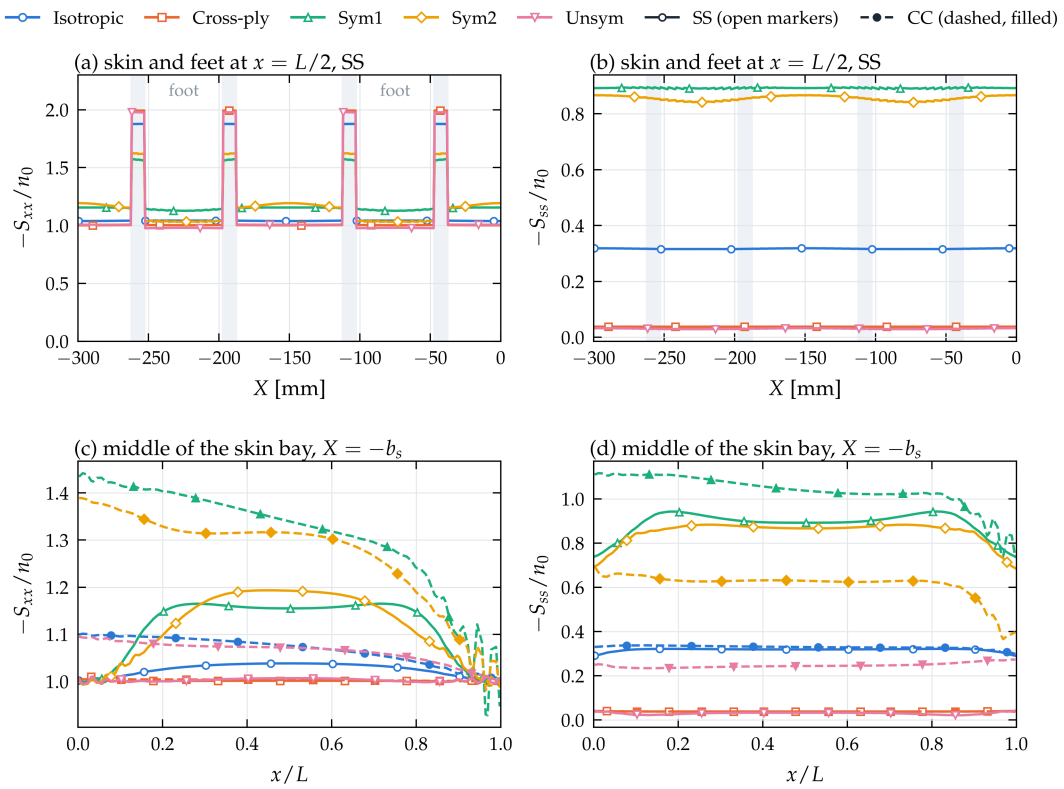


Figure 3. Pre-buckling membrane forces per unit load n_0 (compression positive), $b_s = 150$ mm, $\beta = 1.5$: (a) axial and (b) transverse force at mid-length, SS (shaded: feet); (c) axial and (d) transverse force along the middle of the skin bay; solid/open: SS, dashed/filled: CC.

Table 5. Critical loads N_{cr} [N/mm] of the periodic cells: Abaqus, GBT and the deviation $e = (N_{cr}^{\text{GBT}} - N_{cr}^{\text{Abaqus}})/N_{cr}^{\text{Abaqus}}$.

Lay-up	b_s [mm]	β	SS: Abaqus	SS: GBT	SS: e [%]	CC: Abaqus	CC: GBT	CC: e [%]
Isotropic	150	0.75	127.650	126.284	−1.07	146.600	145.225	−0.94
	150	1.5	123.880	122.674	−0.97	126.100	124.801	−1.03
	150	3	122.740	121.412	−1.08	119.420	118.133	−1.08
	285.5	0.75	23.979	23.954	−0.10	30.678	30.646	−0.11
	285.5	1.5	23.720	23.707	−0.05	24.713	24.675	−0.15
	285.5	3	22.873	22.852	−0.09	22.913	22.870	−0.19
Cross-ply	150	0.75	34.848	34.530	−0.91	49.330	48.878	−0.92
	150	1.5	31.464	31.243	−0.70	36.806	36.503	−0.82
	150	3	31.061	30.810	−0.81	32.275	32.027	−0.77
	285.5	0.75	6.607	6.595	−0.18	11.250	11.228	−0.19
	285.5	1.5	6.609	6.597	−0.19	7.749	7.737	−0.16
	285.5	3	6.304	6.296	−0.13	6.709	6.699	−0.16
Sym1	150	0.75	22.880	22.675	−0.90	24.199	24.047	−0.63
	150	1.5	17.475	17.315	−0.92	15.825	15.606	−1.38
	150	3	15.202	15.078	−0.82	13.077	12.882	−1.49
	285.5	0.75	4.760	4.762	+0.03	5.572	5.564	−0.15
	285.5	1.5	4.011	4.004	−0.18	3.849	3.813	−0.94
	285.5	3	3.832	3.819	−0.33	3.501	3.472	−0.84
Sym2	150	0.75	24.291	24.024	−1.10	30.485	30.252	−0.76
	150	1.5	21.386	21.118	−1.25	23.877	23.563	−1.31

Lay-up	b_s [mm]	β	SS: Abaqus	SS: GBT	SS: e [%]	CC: Abaqus	CC: GBT	CC: e [%]
	150	3	20.415	20.174	−1.18	21.954	21.619	−1.52
	285.5	0.75	5.033	5.026	−0.14	6.679	6.683	+0.06
	285.5	1.5	4.758	4.744	−0.29	5.021	5.005	−0.32
	285.5	3	4.620	4.602	−0.39	4.557	4.530	−0.59
Unsym	150	0.75	33.489	33.277	−0.63	36.370	36.129	−0.66
	150	1.5	32.561	32.260	−0.92	30.238	30.030	−0.69
	150	3	31.825	31.616	−0.66	28.027	27.821	−0.73
	285.5	0.75	6.775	6.806	+0.46	7.707	7.718	+0.14
	285.5	1.5	5.880	5.897	+0.28	6.002	6.013	+0.19
	285.5	3	5.864	5.876	+0.21	5.446	5.455	+0.17

The deviations depend mainly on the pitch: for $b_s = 285.5$ mm the SS mean deviation is 0.21 % (maximum 0.46 %), for $b_s = 150$ mm 0.93 % (maximum 1.25 %), with all 15 GBT results below Abaqus. This offset is not a truncation error (the complete basis is used), not a discretisation error of GBT (finer segments lower the GBT result further, Table 6) and not caused by flexible link walls (stiffer links leave it unchanged, Section 3.7). It is most likely due to the different modelling of the stringer–skin junction (one combined foot laminate with link walls in GBT, two shells tied by *TIE in Abaqus), which governs a larger part of the buckle in the short skin bays of the narrow pitch; closed-form models of the same panels also deviate most from shell models for $b_s = 150$ mm [18].

3.4. Buckling modes and mode decomposition

All short cells buckle locally. For the Unsym cell with $b_s = 150$ mm and $\beta = 1.5$ (Figure 6), with MAC = 0.9998 (SS) and 0.9999 (CC) between GBT and Abaqus, the skin bays buckle in three skewed half-waves (SS) or two hardly skewed ones (CC), and the stringers move only slightly. Over all short cells, Eq. (41) assigns 96–101 % to the local family, up to 3.5 % to the global family (cross-ply cell with $b_s = 285.5$ mm) and −1.6 % to 1.1 % to the distortional family; T and S contribute less than 0.1 % (Figure 7a–c). Values slightly outside 0–100 % stem from coupling energies, which can be negative.

The hierarchical modes (Section 2.2.5) give a finer decomposition, in the manner of classical GBT (Figure 7d–f). For the Unsym cell with SS edges, computed with the complete basis in hierarchical form (N_{cr} = 32.26 N/mm, relative difference from the nodal basis 1.3×10^{-10}), the four modes 12, 18, 8 and 20 carry 96.5 % of the strain energy and the first 20 modes 97.9 %.

3.5. Long cells: change of the buckling mode

Figure 8 extends the SS cells with $b_s = 150$ mm to $\beta = 16$ ($L = 2400$ mm), and Figure 9 shows typical modes. The change of the mode depends on A_{12}/A_{11} of the skin, which sets S_{ss} (Section 3.2).

The Cross-ply and Unsym cells (small S_{ss}) buckle locally up to $\beta = 10$ (30.61 and 31.44 N/mm) and globally from $\beta = 12$, with 99.9 % global participation (Figure 9c); the load then decreases as for an Euler column, for the Unsym cell from 22.41 N/mm at $\beta = 12$ to 12.71 N/mm at $\beta = 16$ (ratio 1.76 against $(16/12)^2 = 1.78$), and local buckling becomes the second eigenvalue (31.43 N/mm).

The isotropic and angle-ply cells switch from $\beta \approx 6$ to a mode with a much lower, decreasing critical load, e.g. for Sym1 14.69, 5.20 and 4.40 N/mm at $\beta = 4$, 8 and 16. The skin forms one full wave across the cell and one half-wave along it, adjacent stringers deflecting oppositely (Figure 9a; about 85 % local, 15 % distortional). This mode has the period $2b_s$ ($k_B b_s = \pi$, Section 2.2.5) and would be missed by a cell of one pitch. It is driven by S_{ss}: without S_{ss} in $\mathbf{G}$ (dashed lines in Figure 8a), the critical load of the isotropic cell with $\beta = 8$ rises from 75.09 to 148.72 N/mm with a local mode (Figure 9b), and the Sym1 cell buckles globally at 17.73 instead of 5.20 N/mm. As for a plate under transverse compression, the load decreases with the length until the global mode, which is hardly affected by S_{ss} (72.28 against 72.33 N/mm at $\beta = 12$), takes over at $\beta = 14$ for the isotropic cell.

Abaqus confirms both mechanisms: with constant width it gives 75.32 and 5.22 N/mm for the long-wave modes (isotropic, Sym1, $\beta = 8$) and 22.37 and 22.29 N/mm for the global modes (Unsym, $\beta = 12$; Cross-ply, $\beta = 14$), within 0.46 % of GBT and with MAC ≥ 0.9999 (Figures 8a and 9). The long-wave mode is thus

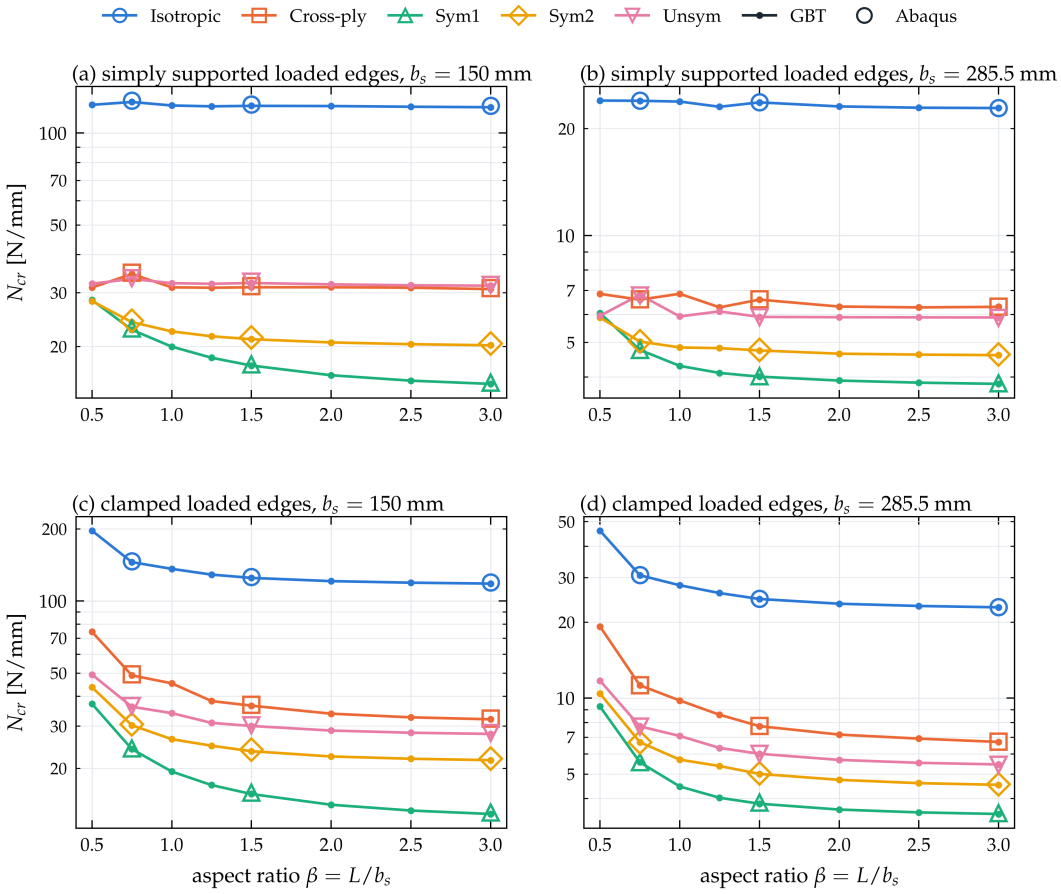


Figure 4. Critical load N_{cr} (logarithmic scale) as a function of the aspect ratio: (a, b) simply supported and (c, d) clamped loaded edges; (a, c) $b_s = 150$ mm and (b, d) $b_s = 285.5$ mm. Lines: GBT; open markers: Abaqus.

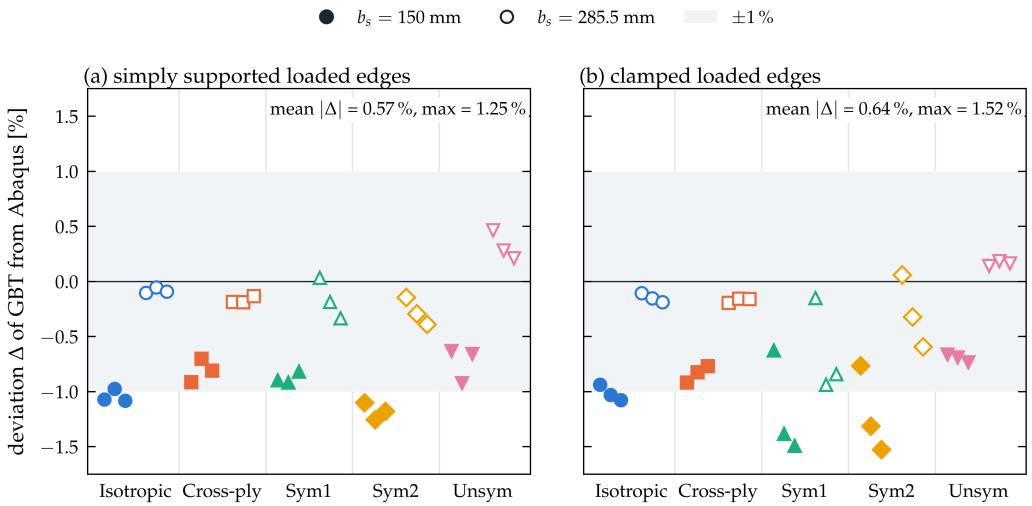


Figure 5. Deviation of the GBT critical loads from Abaqus: (a) simply supported and (b) clamped loaded edges. Filled: $b_s = 150$ mm, open: $b_s = 285.5$ mm; $\beta = 0.75, 1.5, 3$ from left to right; shaded: ± 1 %.

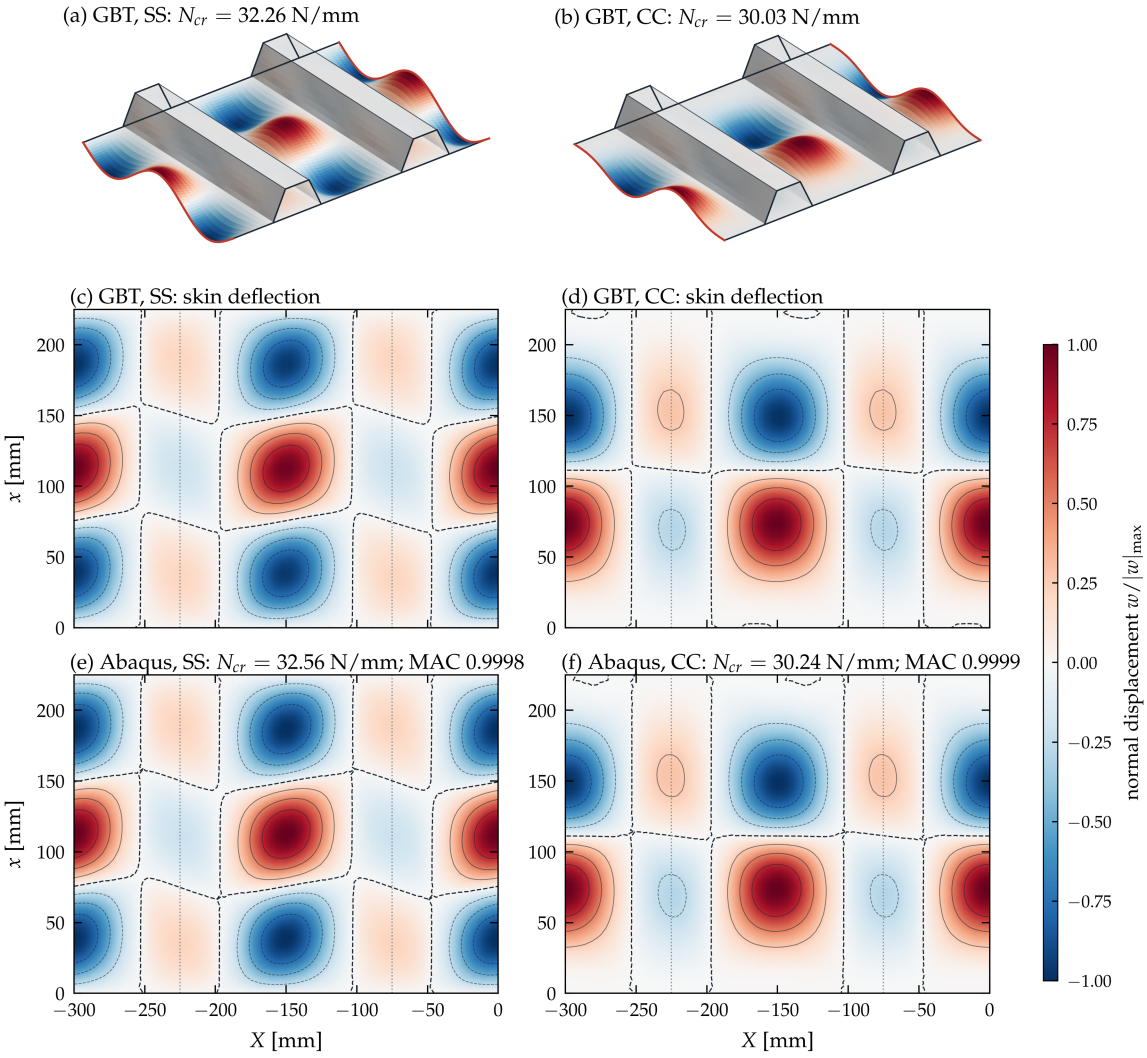


Figure 6. Critical mode of the Unsym cell, $b_s = 150$ mm, $\beta = 1.5$: (a, b) deformed cell from GBT (colour: displacement normal to the walls); skin deflection $w/|w|_{\max}$ from (c, d) GBT and (e, f) Abaqus (contours at ± 0.25, ± 0.5, ± 0.75; dashed: $w = 0$); left: SS, right: CC.

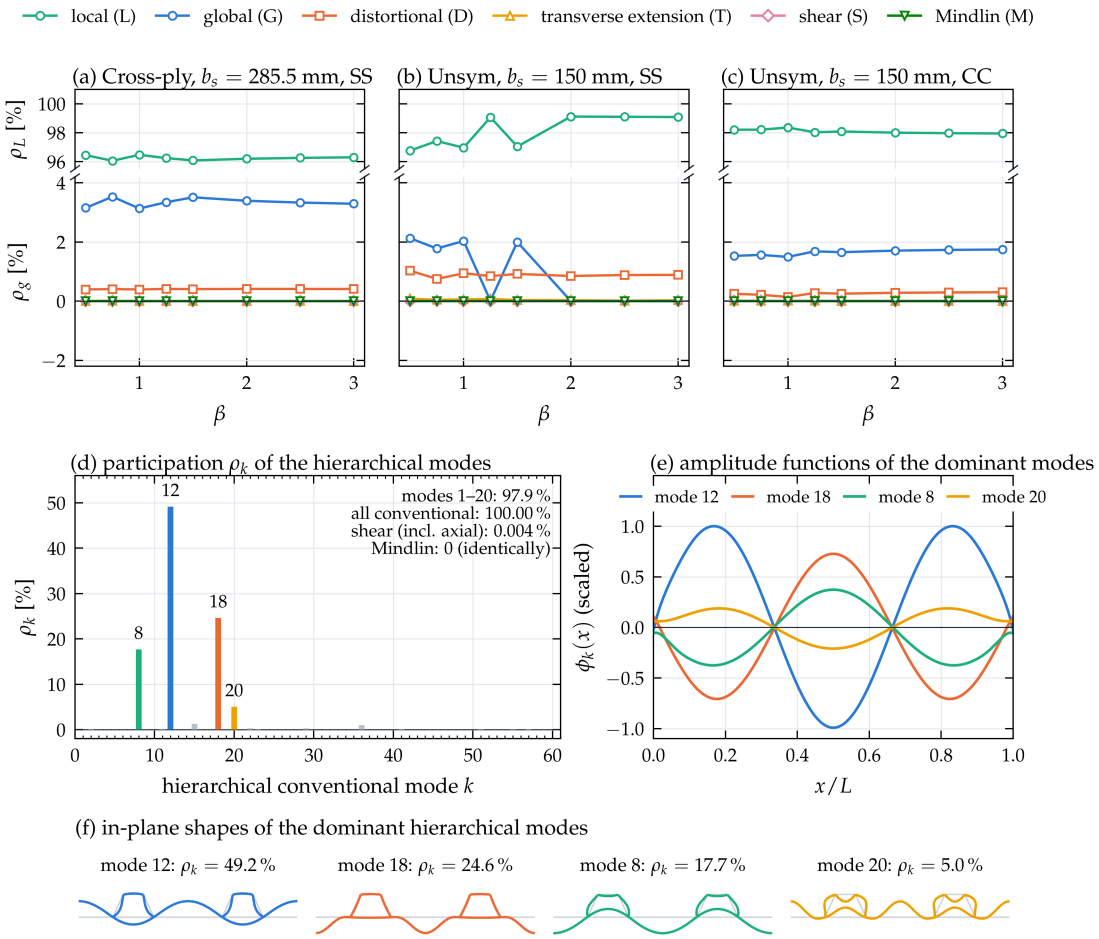


Figure 7. Mode decomposition. (a)–(c) Family participation ρ_g versus β: (a) Cross-ply, $b_s = 285.5$ mm, SS; (b) Unsym, $b_s = 150$ mm, SS; (c) the same, CC. (d) Participation ρ_k of the first 60 hierarchical conventional modes in the critical mode of the Unsym cell, $b_s = 150$ mm, $\beta = 1.5$, SS; (e) amplitude functions and (f) in-plane shapes of the four dominant modes.

caused by the constant cell width, which periodic shell models impose as well. If the shell model lets the cell expand laterally (common jump of $\mathcal{U}_1$ and $\mathcal{U}_3$ between the side edges), the isotropic cell with $\beta = 8$ buckles locally at 152.44 N/mm and the Sym1 cell globally at 15.40 N/mm, i.e. 2.0 and 2.9 times higher than with constant width. Omitting S_{ss} in GBT reproduces the local case within 2.4 % but overestimates the global load of the Sym1 cell by 15 %, probably because the free model also releases the in-plane shear (common jump of $\mathcal{U}_3$); free expansion in GBT requires additional generalised coordinates for these jumps and is left for future work. Constant width, as in periodic shell models, is thus conservative for panels that can expand laterally, whereas omitting S_{ss}, as models that prescribe only the axial load implicitly do, is not conservative in general (Sym1 cell: +15 %).

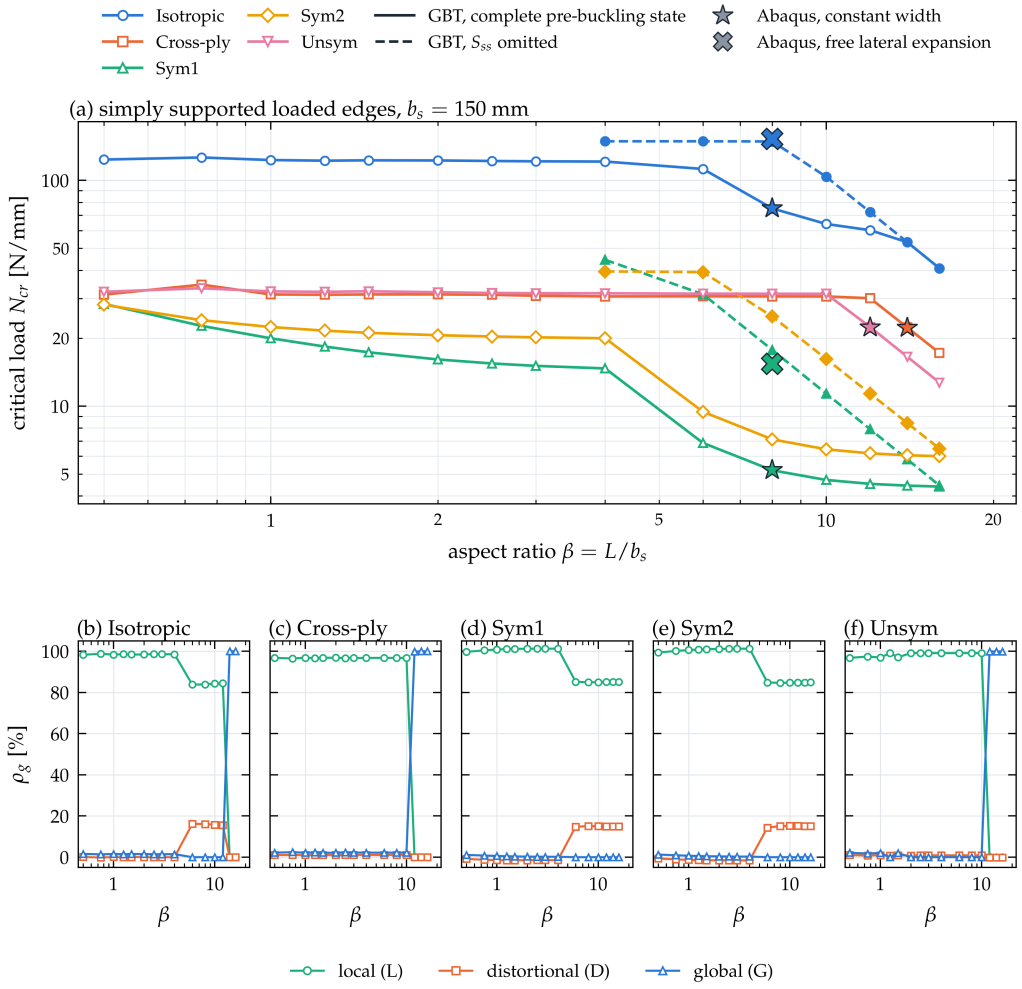


Figure 8. Long cells, SS, $b_s = 150$ mm: (a) critical load versus aspect ratio (logarithmic scales); solid: complete pre-buckling state, dashed: S_{ss} omitted from **G**; stars: Abaqus, constant width; crosses: Abaqus, free lateral expansion; (b)–(f) participation of the local, distortional and global families.

3.6. Hierarchical mode bases

Whether a few modes suffice, as often in GBT, was examined by keeping the first n_C hierarchical conventional modes ($k_r = \pi/b_s$) together with all shear and Mindlin modes, with the pre-buckling state of the complete basis (Figure 10). The convergence is stepwise; the four modes from $n_C = 17$ to 20 reduce the deviation by a factor of 6 to 29. A deviation below 1 % requires 20 to 40 modes for the isotropic, cross-ply and angle-ply cells, but 80 to 100 of the 273 modes for the unsymmetric cell. Omitting the Mindlin modes raises the critical load by 0.36 % to 2.01 %. A classical small basis (20 conventional modes, their warping at k_r, no Mindlin modes; 41 modes) gives +1.45 % for the isotropic cell with $b_s = 285.5$ mm and $\beta = 0.75$, but +10.36 % for the Unsym cell with SS edges.

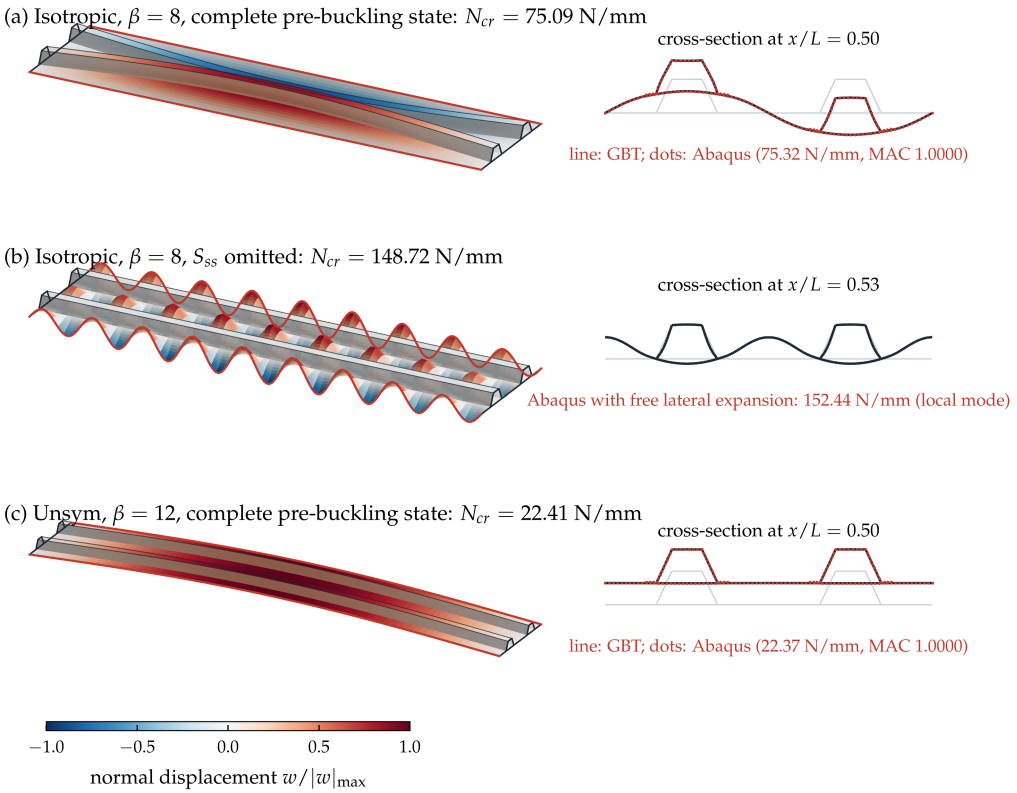


Figure 9. Critical modes of long cells (SS, $b_s = 150$ mm): deformed cell and cross-section at the largest displacement (line: GBT; dots: Abaqus, in (a) and (c)). (a) Isotropic, $\beta = 8$: long-wave mode; (b) the same with S_{ss} omitted: local mode; (c) Unsym, $\beta = 12$: global mode.

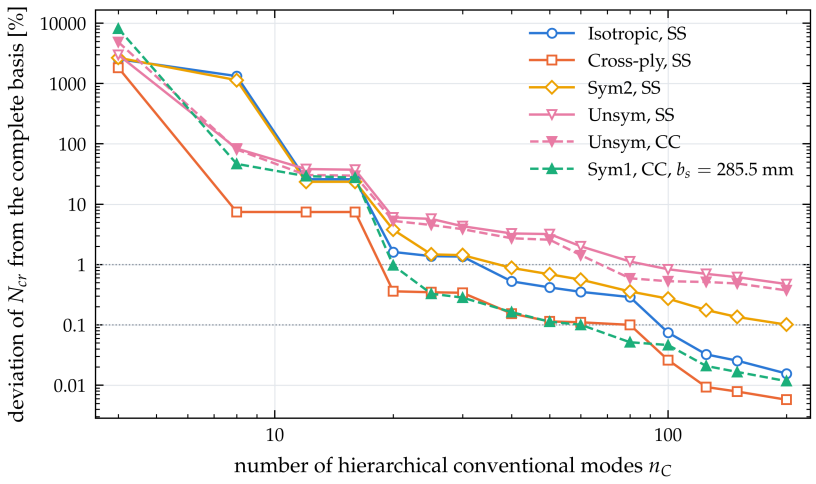


Figure 10. Deviation of the critical load from the complete basis versus the number n_C of hierarchical conventional modes, $\beta = 1.5$.

3.7. Accuracy, efficiency and limitations

The default discretisation (segments of at most $b_s/25$, spans $b_s/20$) was checked by refining each parameter in turn (Table 6). All changes are within 0.50 %, except for the Sym1 cell with CC edges, $b_s = 285.5$ mm and $\beta = 1.5$, for which halving the spans repeatedly raises the critical load by +0.26, +0.47 and +0.62 %, lowering its deviation from Abaqus from −0.94 to −0.32 %. The critical load depends neither on the link-wall factor (122.667, 122.674 and 122.671 N/mm for $k_L = 10^2$, 10^3, 10^4; isotropic cell, SS, $\beta = 1.5$) nor on the soft factor of Eq. (7) (122.684 to 122.673 N/mm for 10^{-2} to 0), whereas a stiff twisting term $D^L_{66} = k_L D_{0,66}$ would lock the web roots (+22.8 %).

Table 6. N_{cr} [N/mm] with the default and refined discretisations (change in brackets).

Case	Default	Spans $h_x/2$	Segments $b_s/40$
Isotropic, SS, $b_s = 150$, $\beta = 1.5$	122.6740	122.5921 (−0.07 %)	122.4080 (−0.22 %)
Sym2, SS, $b_s = 150$, $\beta = 3$	20.1744	20.1776 (+0.02 %)	20.1476 (−0.13 %)
Unsym, CC, $b_s = 150$, $\beta = 1.5$	30.0295	30.0275 (−0.01 %)	29.8795 (−0.50 %)
Sym1, CC, $b_s = 285.5$, $\beta = 3$	3.4715	3.4780 (+0.19 %)	3.4698 (−0.05 %)
Sym1, CC, $b_s = 285.5$, $\beta = 1.5$	3.8130	3.8230 (+0.26 %)	3.8100 (−0.08 %)

A complete GBT analysis with 474–558 modes takes 4.6 s on average for the ten cells with $b_s = 150$ mm and $\beta = 1.5$, of which the cross-section analysis, reused for all lengths, takes less than 1 s. For the four long cells of Table 7, GBT has about 32 % of the degrees of freedom of the shell model and needs 24–51 % of its time (different two-core machines); since the global and distortional modes couple with all others, the GBT matrices are less sparse, and the main benefit is the modal description.

Table 7. Model size and computing time of the long cells (SS, $b_s = 150$ mm; Abaqus: 8 eigenvalues, time of the solver step, Intel i5-13420H; GBT: 2 eigenvalues and mode decomposition, Intel Xeon 2.1 GHz; 2 cores each).

Cell	Abaqus DOFs	Abaqus time [s]	GBT unknowns	GBT time [s]
Isotropic, $\beta = 8$	282,750	167	90,770	61
Sym1, $\beta = 8$	282,750	146	90,770	35
Unsym, $\beta = 12$	423,690	492	135,410	132
Cross-ply, $\beta = 14$	495,030	208	157,730	106

The plate assumptions (Reissner–Mindlin walls, link walls for the tie, linear buckling about a first-order state) are the likely origin of the −1 % offset for $b_s = 150$ mm, and the formulation represents panels whose lateral expansion is restrained (Section 3.5).

4. Conclusions

Periodic boundary conditions were built into the cross-section analysis of shear-deformable GBT: conventional modes by restricting each mode family to its periodic subspace, the warping by a constrained least-squares problem on the cell as a closed contour, and shear and Mindlin modes by merging the side-node functions. Every mode, and hence every truncated basis, is exactly periodic, and the formulation reduces to classical GBT (confirmed to 0.001 %).

For 30 omega-stringer cells with simply supported loaded edges, the GBT–Ritz critical loads agree with Abaqus shell models within 0.57 % on average and 1.25 % at most, and with clamped loaded edges within 0.64 % and 1.52 %, for all lay-ups, with mode shapes agreeing to MAC > 0.999. Cells up to three stringer pitches long buckle locally. From about six pitches, isotropic and angle-ply cells buckle in a long-wave mode driven by the transverse compression that the constant cell width induces, and from twelve pitches cross-ply

and unsymmetric cells buckle globally; with free lateral expansion, Abaqus gives 2.0–2.9 times higher critical loads for the two cells examined. Since periodic shell models impose the same constant width, this must be considered for panels whose lateral expansion is not restrained.

Future work will address periodic conditions with free lateral expansion, enriched bases for coupled laminates, the comparison with the closed-form solutions of El Yaakoubi-Mesbah and Mittelstedt [17], and the use of the formulation as a data generator for machine-learning surrogate models [22].

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